QueryMaster: An End-to-End Text-to-SQL Autonomous Agent

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***Abstract*—This work introduces QueryMaster, an autonomous text-to-SQL agent that automates database queries by utilizing a quantized Large Language Model (LLM). In particular, Low-Rank Adaptation (LoRA) designed for Apple Silicon via the MLX framework is used to refine the Qwen3-8B-4bit model. The model is trained using a reliable combination of the BIRD and SynSQL datasets to guarantee mastery over cross-domain schemas and sophisticated SQL operations. The suggested solution achieves a competitive execution accuracy of 67.4% after being thoroughly tested on the Spider 1.0 development set. Additionally, the model's practical feasibility is successfully demonstrated through the development of a lightweight web interface for real-time local inference. The findings show that highly proficient text-to-SQL agents may be produced by parameter-efficient fine-tuning on consumer-grade hardware, setting the stage for future scalable microservices integration.**

***Index Terms*— Text-to-SQL, Large Language Models, Low-Rank Adaptation, Database Automation.**

I. Introduction

IT is usually necessary to be proficient in Structured Query Language (SQL) in order to access and analyze data in relational databases. For non-technical users, this causes a major slowdown. Promising outcomes in converting natural language queries into executable SQL queries (Text-to-SQL) have been demonstrated by recent developments in Large Language Models (LLMs). However, a lot of processing power is needed to implement these complex models. This project presents QueryMaster, a self-sufficient text-to-SQL agent built to function well on consumer-grade hardware. Without using cloud-based APIs, QueryMaster seeks to close the gap between intricate database structures and natural language comprehension through the use of hardware-specific optimizations and parameter-efficient fine-tuning approaches.

II. Methodology

A. Data Engineering

A unique dataset was created by combining the BIRD and SynSQL datasets in order to train a reliable model that could handle a variety of database systems. This combination exposes the model to cross-domain schemas, nested subqueries, and intricate JOIN procedures. A ChatML format appropriate for conversational instruction tweaking was created from the raw data.

B. Model and Hardware Optimization

The base model selected for this project is Qwen3-8B, quantized to 4-bit precision. To overcome the memory constraints of local training, Low-Rank Adaptation (LoRA) was implemented. The training pipeline was specifically optimized for Apple Silicon (Unified Memory Architecture) using the MLX framework, preventing Out-Of-Memory (OOM) errors and significantly accelerating the training speed.

C. Prototype Interface

For the midterm demonstration, a lightweight web application was developed using the Streamlit framework. This interface allows users to input their database schema (DDL) and natural language queries, enabling real-time, local inference testing of the fine-tuned adapters.

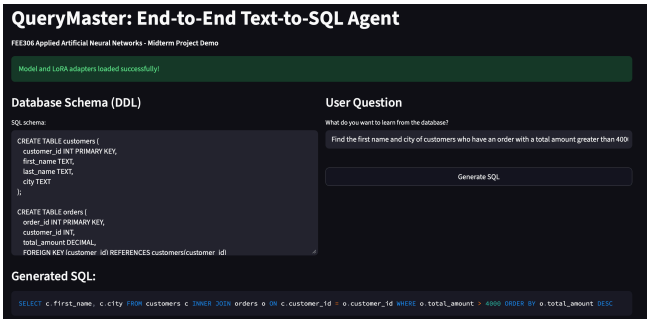

Figure 1: Streamlit web interface for real-time inference.

III. Results

A. Execution Accuracy

The fine-tuned model was evaluated on the highly competitive Spider 1.0 development set, which contains complex schema layouts and difficult query structures. The model achieved an overall Execution Accuracy of 67.4% (697 out of 1034 queries matched successfully). Additionally, the model reduced syntax errors to just 4.1% while handling schema-heavy prompts with great resilience.

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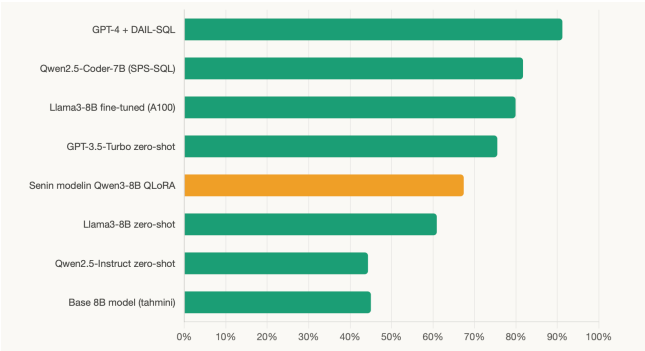

Figure 2: Execution accuracy comparison against baseline LLMs.

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(base) mehmetzahitavci@Mehmet-MacBook-Pro QueryMaster %
============================================================
EVALUATION RESULTS (Spider 1.0 Dev Set)
============================================================
  Execution Accuracy : 67.4% (697/1034)
  SQL Syntax Errors  : 42 (4.1%)

  Accuracy by Difficulty:
    Easy        :  80.4% (344/428)
    Medium      :  63.2% (277/438)
    Hard        :  44.0% (51/116)
    Extra Hard  :  48.1% (25/52)
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Figure 3: Execution accuracy breakdown by query difficulty.

B. Inference Performance

Real-time applicability is an important statistic for autonomous agents that goes beyond accuracy. The model demonstrated an amazing generation speed of roughly 40–45 tokens per second during local inference testing using the MLX framework. The system's effectiveness for real-world production contexts was demonstrated by the generation of complicated SQL queries involving INNER JOINs in less than 1.5 seconds.

IV. Conclusion and Future Work

This midterm project effectively showed the feasibility of implementing an autonomous text-to-SQL agent on consumer-grade hardware. The Qwen3-8B model obtained a highly competitive Execution Accuracy of 67.4% on the Spider assessment suite by utilizing the MLX framework and LoRA fine-tuning. The findings show that a robust combination of training datasets (BIRD and SynSQL) and parameter-efficient fine-tuning enable the model to generalize well to complicated, unknown database designs. The created Streamlit prototype effectively verifies the model's capacity for real-time inference.

The ultimate objective of QueryMaster is to function as an enterprise-grade application, even though the current version offers a strong proof-of-concept. Future research will therefore concentrate on converting this design into a microservices pipeline that is completely scalable. A FastAPI service will contain the AI inference engine. To manage safe database executions, a main backend powered by Java and Spring Boot will then be incorporated. Lastly, the existing prototype will be replaced by a dynamic React frontend that will enable non-technical users to see the generated SQL outputs using automated graphs and charts.

References

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